

Electrical Symbols

IEC 60617 & ANSI/IEEE Complete Reference

Passive Components • Switches • Protection • Designators • Schematics

What You Will Learn

- IEC 60617 vs ANSI/IEEE symbols — how they differ and when each is used
- Passive components: resistor, capacitor, inductor, battery, earth, AC/DC sources
- Contacts and switches: NO, NC, pushbutton, E-stop, selector, motor symbols
- Reference designator system (IEC 81346) and how to read a complete schematic

How to Read a Reference Designator

KM1 — K = contactor, M = main, 1 = number

S0 = E-stop | S1 = start PB | F1 = fuse | KA1 = relay | M1 = motor

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ELEC-008: Electrical Symbols

Electrical symbols are the universal language of circuit diagrams and schematic drawings. Just as engineers worldwide can read a structural drawing without speaking the same language, a well-drawn electrical schematic using standard symbols communicates the circuit's function to any trained electrical professional. Knowing the symbols — and the standard they belong to — is a prerequisite for reading and drawing any professional electrical document.

Learning Objectives

- Identify the IEC 60617 and ANSI/IEEE symbol standards and know when each is used
- Recognise and draw the fundamental electrical symbols for passive components and sources
- Read and draw switch, contact, and protection device symbols correctly
- Use the IEC 81346 reference designator system to label all components
- Follow a systematic process to read any unfamiliar electrical schematic diagram

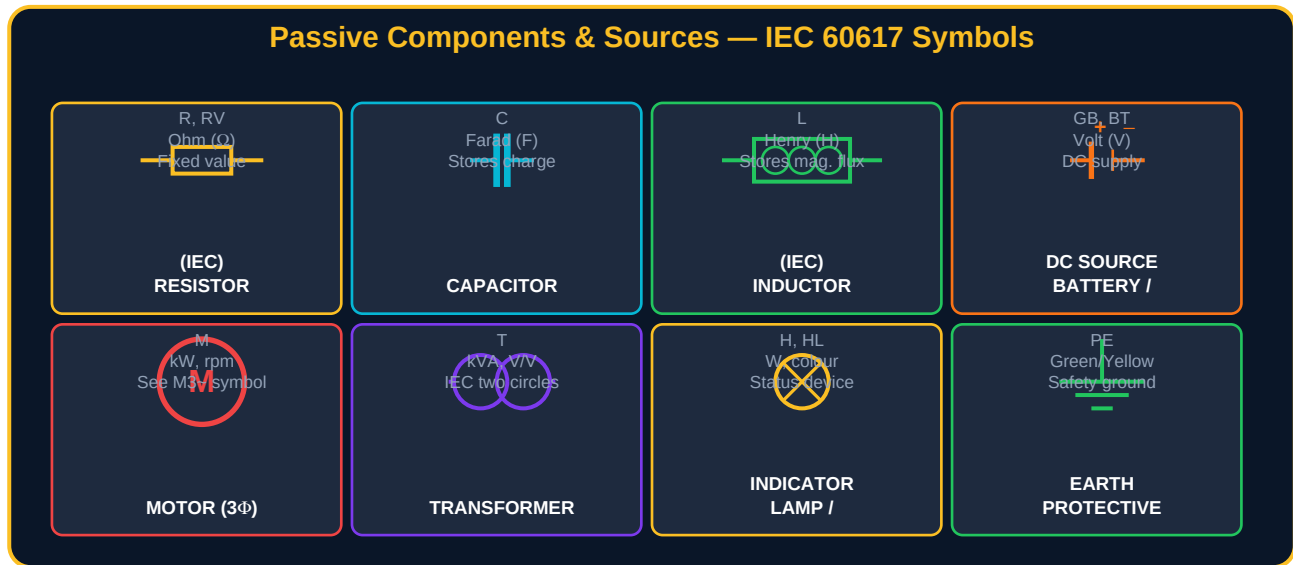
1. Symbol Standards — IEC 60617 vs ANSI/IEEE

Two major symbol standards exist worldwide. The standard used on a drawing is usually stated in the title block. When it is not, use context clues: if the resistor looks like a rectangle, it is IEC; if it looks like a zigzag, it is ANSI. Both convey the same circuit information — only the visual representation differs.

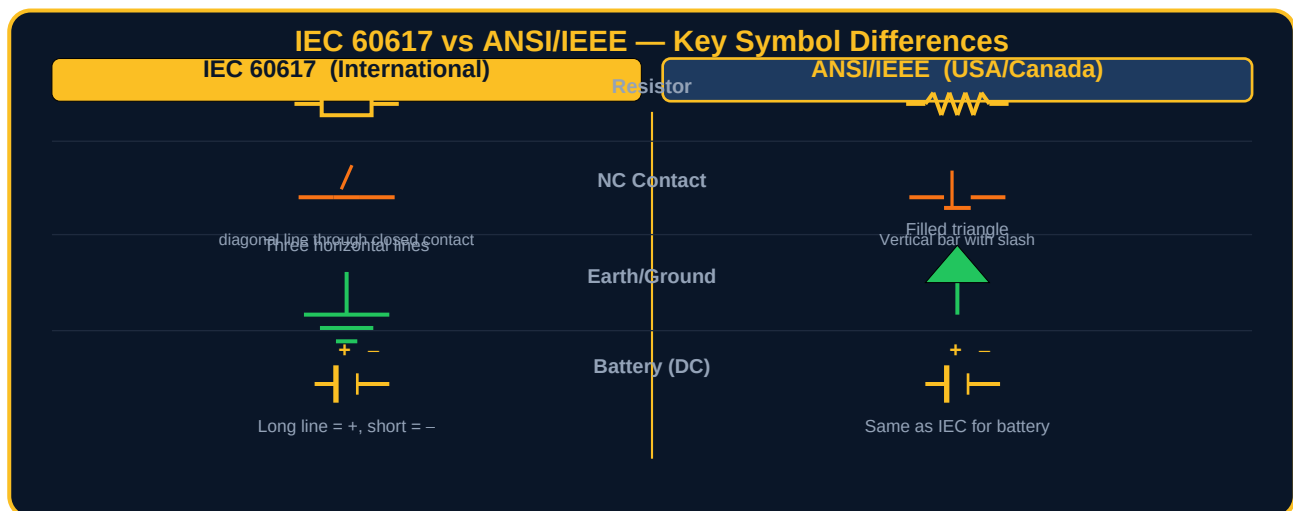
Aspect	IEC 60617 (International)	ANSI/IEEE (American)
Used in	Europe, Africa, Asia, Middle East, South America, most of the world	USA, Canada. Also seen on equipment exported from USA manufacturers.
Resistor symbol	Rectangle (box) on a wire	Zigzag (sawtooth) wave on a wire
Earth/Ground	Three horizontal lines, decreasing in length	Filled triangle pointing downward
NC contact	Closed bridge line with diagonal slash through actuator	Two vertical bars with diagonal slash — contact shown as a break in the line
Fuse	Rectangle with a line through the centre	S-shaped or wavy line inside a rectangle
Power quality	Higher precision — symbols fully defined in standard	More variation between manufacturers — some symbols differ by company

When to use	Any project outside North America. Default for IEC-standard equipment (most industrial equipment globally).	Projects in North America, or when working with American equipment datasheets and drawings.
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2. Passive Components and Power Sources



IEC vs ANSI Symbol Comparison



3. Switch and Contact Symbols

Switch & Contact Symbols IEC 60617

NO CONTACT



S, K-NO
when coil ON
Passes current
Normally Open

NC CONTACT



S, K-NC
when coil ON
Breaks current
Normally Closed

PUSHBUTTON (NO)



SB-NO
Spring returns
Momentary action
Start, Alarm Reset

MOTOR (M3~)



M
kW, rpm, V
M3~ = 3-phase
3-Phase motor

FUSE



F, FU
Fixed rating
protection
Short-circuit

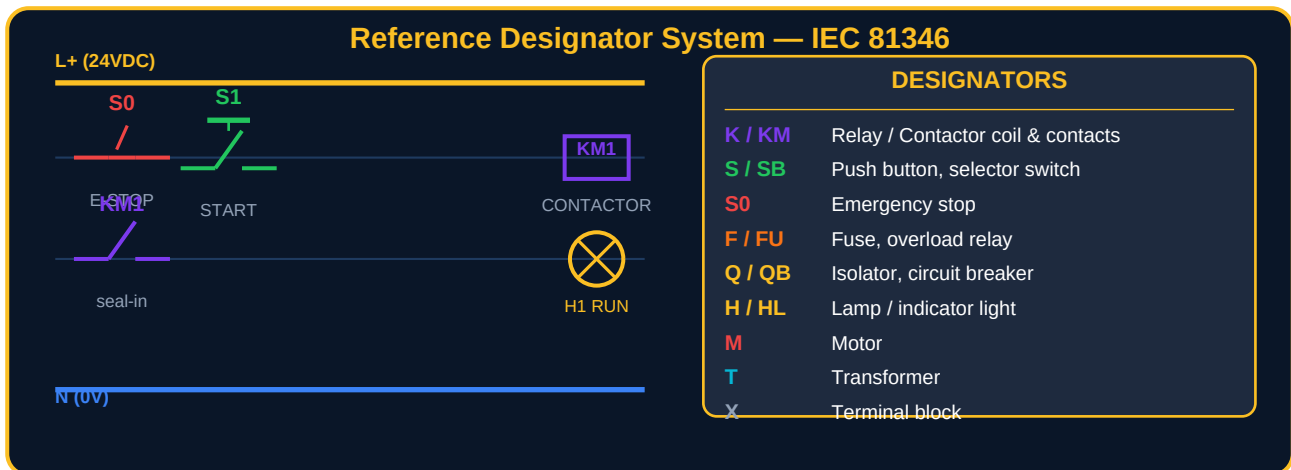
Symbol	IEC Description	Function	Common Application
Normally Open (NO) contact	Open bridge on angled actuator line	Circuit is OPEN when unactuated. Closes when coil energises or button pressed.	Start pushbuttons, seal-in contacts, motor run contacts (KM1-NO)
Normally Closed (NC) contact	Closed bridge with diagonal slash through actuator	Circuit is CLOSED when unactuated. Opens when coil energises.	Stop pushbuttons, E-stop contacts, overload trip contacts (FA1-NC)
Pushbutton NO (S-NO)	NO contact with manual actuator (arrow/bar)	Momentary NO contact — spring returns after release	START button (green), reset buttons, test buttons
Pushbutton NC (S-NC)	NC contact with manual actuator	Momentary NC contact — spring returns after release	STOP button (black/red), E-stop button (red/yellow, twist-release)
Selector switch (2-position)	NO or NC contact with rotary actuator	Maintained position — stays in position when turned	AUTO/MANUAL selector, LOCAL/REMOTE, ON/OFF key switch
Float switch	NO or NC contact with float actuator symbol	Operated by liquid level	Tank high/low level alarm, pump start on low level

Pressure switch	NO or NC with "P" or pressure actuator	Operates when pressure reaches set point	Pump overload protection, air compressor pressure control
Limit switch (position)	NO or NC with roller actuator	Mechanical actuation by moving machine part	End-of-travel detection on conveyors, gates, actuators

4. Protection Device Symbols

Device	IEC Symbol Description	Function	IEC Reference
Fuse (F)	Rectangle with a straight line through the centre	Sacrificial device — melts on overcurrent to protect downstream circuit	F, FU (older UK)
MCB / MCCB (Q)	Box with break symbol inside + toggle actuator	Automatically trips on overcurrent and short circuit. Can be reset without replacement.	Q, QB, QM (motor protection)
RCCB / RCD (Q or F)	MCB symbol with test button + "I _{Δn} " annotation	Trips on earth leakage current. Protects against electric shock (30mA) or fire (300mA).	Q, F (depends on standard in use)
Overload relay (FA)	Contact symbol with thermal element symbol (bimetal)	Trips if motor draws excess current for sustained period. Manual or auto reset.	FA (overload contactor auxiliary), F
Circuit breaker with earth fault (Q)	MCB symbol with earth fault indicator	Combined short-circuit + earth fault protection in one device	Q, QB
Surge arrester	Triangle pointing to a ground bar	Limits voltage spikes from lightning or switching to protect sensitive equipment	F, V (voltage protection device)

5. Reference Designators (IEC 81346)



The reference designator system gives every component on a schematic a unique, meaningful label. The label has two parts: a **letter code** (identifies the type of component) and a **number** (distinguishes multiple components of the same type). Components of the same type are numbered sequentially: K1, K2, K3.

6. How to Read an Electrical Schematic — 8-Step Process

Step	Action	What to Look For
1	Read the title block	Project name, drawing number, standard (IEC/ANSI), date, revision, drawn by. Establishes context.
2	Find the legend/symbol list	Sheet 1 or separate legend drawing. Identifies any non-standard symbols used on the project.
3	Identify power supply rails	Top and bottom rails of ladder diagrams (or left/right). L+ and N (DC) or L1/L2/L3 and N (AC).
4	Trace the power circuit first	Follow main power from isolator through protection device to load (motor, heater, etc.).
5	Then read control circuits rung by rung	Each rung is one control statement. Read left to right: contacts (conditions) → coil (output).
6	Follow cross-references	When you see K1 coil: find all K1 contacts via the cross-reference. Follow the logic chain.
7	Check earthing	Every circuit should show PE (protective earth) connections. Note isolation and earth paths.
8	Use the terminal strip/block list	Find which terminals connect to field devices. Terminals (X1:1, X1:2...) link internal to external wiring.

7. Practice Exercises

#	Exercise	Task
1	Symbol identification	Sketch the IEC 60617 symbols for: (a) resistor, (b) capacitor, (c) NC contact, (d) fuse, (e) motor (3-phase), (f) transformer, (g) protective earth. Then write the standard reference designator for each.
2	IEC vs ANSI comparison	For each of these symbols, describe the visual difference between IEC and ANSI versions: (a) resistor, (b) earth/ground, (c) NC contact. How would you identify which standard a drawing uses without checking the title block?
3	Ladder rung reading	A rung shows: NC contact labelled S0 → NC contact labelled FA1 → NO contact labelled S1 → coil labelled KM1. In plain English: describe what this rung does and under what conditions KM1 energises.
4	Reference designator assignment	A control panel contains: 3 contactors, 1 auxiliary relay, 2 fuses, 1 transformer, 4 pushbuttons (2 start, 2 stop), 1 E-stop, 3 indicator lamps (run/stop/fault). Assign a complete reference designator to every component.

5	Draw a motor start/stop schematic	Draw the IEC schematic for a DOL motor starter with: E-stop (S0 NC), Stop PB (S1 NC), Start PB (S2 NO), KM1 seal-in (KM1-NO in parallel with S2), KM1 coil, Run lamp (H1 connected to KM1-NO), and Fault lamp (H2 connected to FA1-NC overload contact).
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Lesson Summary

Key Point	Summary
Two standards	IEC 60617 = international (rectangle resistor, three-line earth). ANSI/IEEE = USA/Canada (zigzag resistor, triangle earth). Both valid — check title block.
NO vs NC contacts	NO (Normally Open) = OPEN at rest, closes when actuated. NC (Normally Closed) = CLOSED at rest, opens when actuated. The name refers to the rest state.
Pushbuttons	NO pushbutton (S-NO): for START functions — momentary contact, spring return. NC pushbutton (S-NC): for STOP and E-STOP — break contact in coil circuit.
Protection devices	Fuse = sacrificial, must replace after operation. MCB = resettable, protects against overload and short circuit. RCCB/RCD = protects against earth leakage (shock/fire risk).
Reference designators	K = contactor/relay. S = switch/pushbutton (S0 = E-stop). F = fuse/protection. Q = CB/isolator. H = lamp/indicator. M = motor. T = transformer. X = terminal block.
Reading schematics	8-step process: title block → legend → power rails → power circuit → control rungs → cross-references → earthing → terminal list.

What Is Next — ELEC-009

ELEC-009: Circuit Design Basics — Introduction to designing electrical circuits from scratch: identifying loads, selecting protective devices, sizing cables, and drawing the complete single-line and schematic diagrams. Series vs parallel circuit analysis. Ohm's Law and power calculations applied to real circuit design decisions.

Resource	Details
ELEC PDF Series	ELEC-001 through ELEC-030 — ayetechub.com/pdfs.html
Wiring Diagrams eBook	Free interactive ebook — ayetechub.com/wiring-diagram-flipbook.html
Community	Telegram: t.me/ayetechub